Skin Color Measurement from Dermatoscopic Images: An Evaluation on a Synthetic Dataset

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***Abstract*—This paper presents a comprehensive evaluation of skin color measurement methods from dermatoscopic images using a synthetic dataset (S-SYNTH) with controlled ground-truth melanosome fractions. Our dataset comprises 10,000 images rendered with varying melanin content, lesion shapes, hair models, and 18 distinct lighting conditions, allowing us to rigorously assess the robustness of different approaches. We compare four categories of techniques—segmentation-based, patch-based, color quantization, and neural networks—for estimating the Individual Typology Angle (ITA) and classifying Fitzpatrick types. Our results show that segmentation-based and color quantization methods yield robust, lighting-invariant estimates, whereas patch-based approaches exhibit significant lighting-dependent biases that require calibration. Furthermore, neural network models, particularly when combined with heavy blurring to reduce overfitting, can provide light-invariant Fitzpatrick predictions, although their generalization to real-world images remains unverified. We conclude with practical recommendations for designing fair and reliable skin color estimation methods.**

***Keywords*—AI fairness; biometrics; colorimetry; dermatoscopy**

I. Introduction

Deep neural networks (DNNs) for classifying dermatoscopic images are widely used and are a very active field of research [1, 2]. However, studies on clinical images have shown DNN models can have marked biases against dark-skinned individuals [3, 4]. It is possible that the same biases also extend to dermatoscopic images. However, accurately measuring this bias requires knowing the skin color of the patient in an image. To date, there are no publicly available datasets of dermatoscopic images with associated skin color data. As a result, researchers often estimate skin color directly from the image pixels, commonly relying on colorimetry methods from image pixel values. However, such estimations are highly sensitive to lighting, image composition, and variations in acquisition protocols.

Building on the insights by Kalb et al. [5], the present work aims to systematically evaluate colorimetry-based skin color measurement techniques. Our central contribution is the use of publicly available system for rendering dermatoscopic images called S-SYNTH [6] in which the *ground truth* (GT) melanin content of the skin can be precisely specified during rendering. This allows us to experimentally vary the lighting conditions while keeping all other variables constant.

This is the first paper to systematically evaluate colorimetry methods on a synthetic dataset with ground-truth melanin levels, enabling precise control over lighting conditions. We compare four colorimetry techniques and a neural network classifier: segmentation-based ITA estimation is the most consistent but demands reliable segmentation; color quantization achieves comparable performance without segmentation; and patch-based approaches show substantial, lighting-dependent biases. Meanwhile, a neural network trained via ordinal regression outperforms all pixel-based methods, albeit at the cost of requiring labeled training data and unknown generalization performance. All code and data is available at github.com/marinbenc/dermatoscopy_colorimetry_eval.

II. Background and Related Work

A recent study by Kalb et al. [5] compared four methods of estimating skin colors from dermatoscopic images approaches and identified substantial discrepancies among them, underscoring the potential for erroneous or inconsistent measurements. We extend their work by including other classes of colorimetry methods as well as evaluating the methods on a synthetic dataset with known ground-truth skin color values.

To do so, we adopt two widely used metrics for assessing skin tone: the Fitzpatrick scale (FP) [7] and the Individual Typology Angle (ITA) [8], both frequently employed in skin color bias research [9, 10].

1) Fitzpatrick scale: This categorization divides skin types into six groups based on UV response, from Type I (very fair, always burns) to Type VI (deeply pigmented, never burns). Although labeling Fitzpatrick types from images can be subjective [9], it serves as a useful framework for large-scale bias analyses.

2) Individual Typology Angle: ITA provides a continuous measure of constitutive pigmentation, with higher values corresponding to lighter skin. Image pixels are transformed into CIE-Lab space and the following equation is applied [10]:

$$\mathrm{ITA}(L^*, b^*) = \arctan\Big(\frac{L^* - 50}{b^*}\Big)\,\frac{180}{\pi}, \qquad (1)$$

where L^* denotes lightness and b^* the blue–yellow axis. While convenient, ITA is strongly affected by lighting and image composition, limiting its utility for classification without additional normalization steps.

III. Methods

The synthetic dataset comprises 10,000 dermatoscopic images and segmentation masks generated by varying melanin content, lesion shape, hair models, and 18 distinct lighting conditions, among other factors, following the procedure outlined in [**?**]. The dataset is well-balanced across each of the image generation parameters.

A. *Obtaining the Ground-Truth Fitzpatrick Type Labels*

To obtain ground-truth Fitzpatrick (FP) labels, we use the segmentation masks provided by SSYNTH to remove lesion regions and compute the mean ITA value for each image, discarding outlier pixels. The resulting ITA values are then binned into FP types using the thresholds from [10]. We empirically derive thresholds for binning by computing the mean melanosome fraction within each FP bin and taking the mid-point between each mean as the threshold values for binning.

By defining the synthetic Fitzpatrick labels directly from the melanosome fraction rather than from pixel-based color values, we ensure that these labels remain invariant to the lighting variations in the dataset.

B. *Dermatoscopy Colorimetry Methods*

In this study, we evaluate a representative implementation of four general classes of skin color estimation methods from dermatoscopic images: (*1*) **segmentation-based**, (*2*) **patch-based**, (*3*) **color quantization**, and (*4*) **model-based** approaches. Below, we briefly describe each category, provide related work, and outline the specific implementation details used in our experiments.

1) Segmentation-based: In this class of methods, healthy skin regions are isolated by removing lesions (and sometimes other artifacts). Traditional approaches employ image processing or deep neural networks for segmentation, then compute the mean color (e.g., in CIELAB space) of the remaining pixels, discarding outliers [10]. This can reduce contamination from lesions, but any segmentation biases can propagate to the color estimation. In our work, we use ground-truth (GT) segmentation masks from the synthetic dataset [6]: we discard lesion pixels, compute the median L^* and b^* values of the remaining skin, exclude pixels lying more than one standard deviation away, and then calculate the ITA (Eq. 1) from the final median values.

2) Patch-based: Instead of segmenting lesions, Bevan and Atapour-Abarghouei [4] propose sampling small patches around the image edges, then choosing the "lightest" patch (highest ITA) under the assumption that non-diseased skin is brighter than lesions or blemishes. Kalb et al. [5] follow the same strategy but replace $\arctan$ with $\arctan 2$ when computing ITA. We adopt this approach and linearly calibrate the resulting estimates to match segmentation-based ITA values, as the raw patch-based outputs show a consistent bias under certain lighting conditions and can fail when patients exhibit hypopigmentation or other anomalies.

3) Color quantization: Clustering-based methods sidestep lesion segmentation by quantizing the image to a small set of dominant colors, then selecting the cluster that most likely represents skin [11]. In our implementation, we first apply CLAHE on the L^* channel and use Otsu thresholding of the HSV value channel, followed by morphological expansion, to mask out lesions and other artifacts. We then convert the masked skin regions back to CIELAB and perform k-means clustering, with the optimal number of clusters determined by the kneedle algorithm [12]. From the largest skin cluster, we compute ITA as in Eq. 1.

4) Model-based: Finally, deep learning methods can be trained to directly predict skin color categories. Such models may learn invariances to illumination or image composition but require sufficient labeled data [9]. We implement an ordinal regression framework (CORAL) [13] on top of a VGG11 backbone [14], with Fitzpatrick-type labels derived by binning synthetic melanosome fractions (Eq. **??**). The 10,000 synthetic images are split into 80% training, 10% validation, and 10% testing. We also apply a heavy Gaussian blur to each input before feeding it to the network, encouraging the model to focus on overall color rather than fine textures. After 100 training epochs, we select the best model checkpoint based on validation loss. All reported results are from the test set.

IV. RESULTS

We first compare the three ITA measurement approaches described in Section III-B: segmentation-based, patch-based, and color quantization. As shown in Fig. 1, the patch-based approach exhibits a consistent linear bias relative to the segmentation-based ITA values under most lighting conditions, prompting us to apply an ordinary least squares (OLS) regression to calibrate its estimates.

Our assumption is that a good estimate of ITA would be perfectly correlated with the melanosome fraction used during the synthetic image generation, as ITA should not vary based on lighting or other variables. Thus, to measure the quality of each prediction method, we compute Pearson's

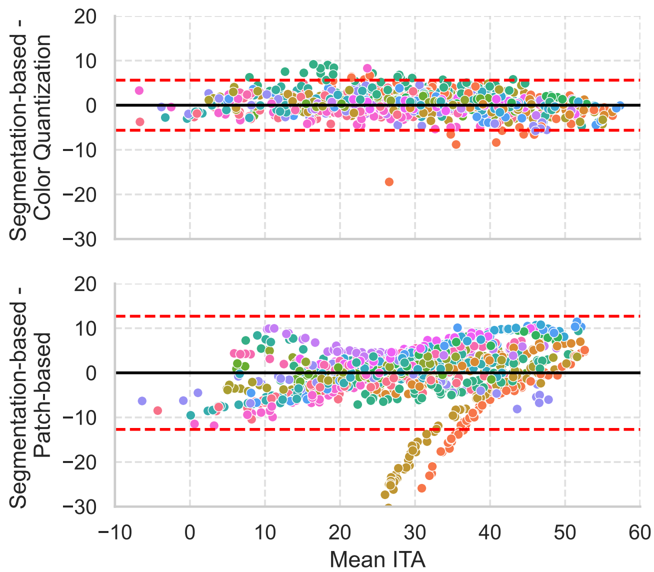

Figure 1. Bland-Altman plots comparing the color quantization and the patch-based to the segmentation-based ITA estimates. Colors indicate different lighting conditions used during image generation, a total of 18 categories.

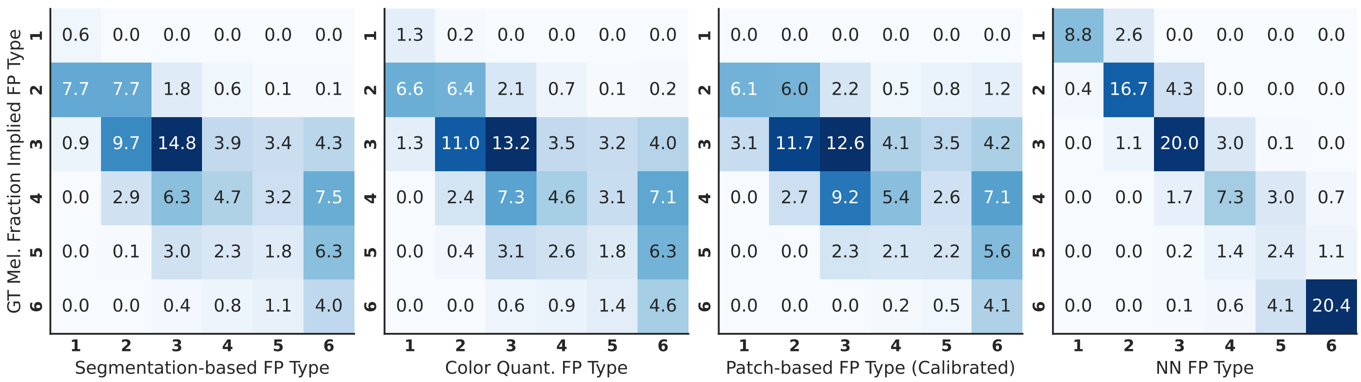

Figure 2. Confusion matrices of Fitzpatrick (FP) type estimates as compared to the ground-truth FP, calculated by binning the melanosome fraction. Numbers indicate % the total number of test images ($n = 1000$).

correlation between each ITA estimate and the ground-truth melanosome fraction, and generate 95% confidence intervals via 1,000 bootstrap resamples. Table I presents the results for the segmentation-based, color quantization, and calibrated patch-based methods.

TABLE I. Pearson's correlation (ρ) between each ITA estimate and ground-truth melanosome fraction, with 95% confidence intervals (CIs) over 1,000 bootstrap resamples.

Method	ρ	**95% CI**
Segmentation-based	0.705	[0.695, 0.714]
Color quantization	0.683	[0.671, 0.693]
Calibrated patch-based	0.624	[0.610, 0.638]

A Bland–Altman analysis (Fig. 1) reveals that the color quantization method agrees more closely with the segmentation-based ITA, whereas the patch-based method exhibits a consistent bias relative to segmentation. This discrepancy is further amplified under certain lighting conditions, even after calibration.

A. Sensitivity to Lighting Conditions

ITA measurements should ideally capture melanin content rather than external illumination. To assess this, we fit two linear models for each estimation method (segmentation-based, color quantization, and calibrated patch-based): ITA $\sim$ mel, and ITA $\sim$ mel + light, where mel is the melanosome fraction used during synthetic image generation, while light is a categorical variable indicating the 18 lighting conditions. We then compute the coefficient of determination (R^2) for both models and measure how much adding light improves the fit (ΔR^2). Table II summarizes these results.

TABLE II. Coefficient of determination R^2 under two linear models of ITA, with and without including lighting conditions. ΔR^2 indicates the difference between the two R^2 values, i.e. the relative sensitivity of each ITA estimate method to lighting conditions.

Method	$ITA \sim mel$	$ITA \sim mel + light$	ΔR^2
Segmentation-based	0.5	0.72	0.23
Color quantization	0.47	0.678	0.21
Calibrated patch-based	0.39	0.7	0.31

From Table II, adding the lighting term increases R^2 for each method, revealing that lighting still exerts a significant influence on ITA estimates. Notably, the calibrated patch-based approach shows the largest ΔR^2 (0.306), indicating a higher sensitivity to illumination compared to the other methods.

B. Fitzpatrick Type Estimation

We evaluate the ability of each method to predict the Fitzpatrick type (FP) of an image, using FP labels based on empirically binning the melanosome fraction values. For the three ITA-based approaches (segmentation-based, patch-based, and color quantization), we convert the estimated ITA values into FP types using the standard thresholds proposed in [10].

Fig. 2 shows the confusion matrices for each of the four methods compared against the ground-truth FP labels. Table III summarizes the balanced accuracy, macro precision, macro recall, macro F1, and quadratically weighted kappa for each approach.

TABLE III. Fitzpatrick Type Estimation Results: Balanced Accuracy (Acc), Macro Precision (Prec), Macro Recall (Rec), Macro F1 (F1), and Quadratically Weighted Kappa (Kappa).

Method	**Acc**	**Prec**	**Rec**	**F1**	**Kappa**
Segmentation-based	0.29	0.46	0.29	0.28	0.59
Color quantization	0.29	0.43	0.29	0.29	0.59
Calibrated patch-based	0.27	0.32	0.27	0.25	0.52
Model-based	0.72	0.71	0.72	0.70	0.95

From Table III, the model-based approach achieves the highest performance on every metric. Among the ITA-based methods, segmentation-based and color quantization exhibit slightly higher balanced accuracy and kappa than the patch-based approach. The confusion matrices (Fig. 2) reveal that errors in the ITA-based methods often involve adjacent FP classes.

V. Conclusion and Discussion

Our results indicate that segmentation-based ITA estimation yields the most robust performance across varied lighting conditions, but this method inherently depends on accurate lesion segmentation—itself potentially prone to bias and error when derived from learned models. By contrast, color quantization avoids the need for segmentation and achieves comparable accuracy and lighting invariance, making it a compelling, interpretable alternative that relies solely on simple clustering rather than complex models.

Patch-based methods exhibit substantial bias tied to individual lighting setups. In practice, each lighting condition appears to introduce a consistent offset that would necessitate calibration if the approach were to be used reliably. Authors proposing new colorimetry techniques should account for these pitfalls and consider calibration strategies for different lighting scenarios.

Our experiments show that neural networks, when combined with heavy blurring to mitigate overfitting on the synthetic dataset, can yield robust light-invariant Fitzpatrick type predictions. However, without large, well-labeled real-world datasets, their generalizability remains unverified, indicating a promising direction for future research.

In summary, we recommend that researchers favor either color quantization or segmentation-based approaches for skin color estimation. These methods have proven more robust and less prone to bias than heuristic strategies, such as selecting the lightest patch, which can systematically introduce errors under varying lighting conditions. We further advise that any colorimetric measurements be calibrated against multiple reference measures and datasets to account for the fixed biases induced by illumination. Finally, neural network approaches show promise for obtaining light-invariant predictions, representing a promising avenue for future research.

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